

Implementation of State-of-Charge and State-of-Health Estimation for Lithium-Ion Batteries

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Abstract—An estimation method of the state-of-charge (SOC) and the state-of-health (SOH) for Li-ion batteries is implemented in this paper. For the proposed methods, we defined the ratio between the variations in voltage and the variations in current in each cell during the operation processes of battery modules as the dynamic impedance, which is used to determine the SOC of the battery. Any initial values are unnecessary in the proposed estimation methods, and the proposed dynamic impedance can better reflect the real electrical characteristics of Li-ion batteries than traditional estimation methods. Furthermore, real-time calculations can be achieved using the proposed methods. The SOH of Li-ion batteries decays as the batteries age, which also influences the value of the dynamic impedance. The projection technique is proposed to determine the SOH based on the dynamic impedance. In this study, a real time platform for estimating SOC and SOH is built, a MCU with a battery management system (BMS) is also integrated. The used BMS receives the related information to estimate the SOC and SOH, and comprises some protection devices to increase battery life spans. Some experimental results are presented to verify the accuracy and feasibility of the proposed theoretical predictions.

I. INTRODUCTION

Because of the global environmental awareness, electric vehicles which do not emit exhaust fumes and cause air pollution [1-2] have gained popularity around the world. Many major automobile manufacturers have developed electric and hybrid vehicles. Nevertheless, these electric or hybrid vehicles have not yet secured a significant share of the market due to high cost. Closer consideration points to the high cost of Li-ion battery, which occupy a significant percentage of the whole costs. Battery modules act as the power source of electric vehicles to determine the endurance ability of the vehicle and some electrical characteristics regarding the vehicle's performance [3-4]. Therefore, for electric vehicles to contend with common gasoline vehicles, a high-quality battery module is essential. However, this in turn increases costs.

Recently, the battery modules used in hybrid and electric vehicles are composed of hundreds battery cells. Even

carefully screened, the electric features of each cell can vary or show differences that become more apparent with time. These differences can then damage the battery module, prevent it from working optimally, or even cause dangerous accidents. To lengthen the life span of battery modules and prevent added costs for consumers, almost battery modules are now incorporated with BMS [5-8]. Although this feature increases costs, its benefits outweigh those costs, making an effective BMS an essential component of a hybrid or electric vehicle. To record voltage, current, and temperature of each cell, the BMS is essential to ensure the safety. We can use BMS to initiate protective functions in the event of abnormalities including overly high or low voltage, discharge currents or temperatures, failures, or short circuits.

To prevent overcharge and over-discharge, the estimation of the state-of-charge (SOC) of batteries is thus a crucial issue [9-10]. Other than SOC, batteries also have another important parameter: state-of-health (SOH), which represents the gauge of aged battery modules. To calculate the SOH is also an important issue to get the accurate aging conditions and available capacity in battery module.

II. DESCRIPTION OF THE PROPOSED ESTIMATION RULES

Recently, the existing SOC estimation methods have some common limitations, including (1) the initial values or historical data are essential, (2) the incapability to execute real-time estimation, (3) the incapability to reflect real SOC from estimated SOC when SOH is changed. Similarly, current SOH estimation methods also possess shortcomings that are difficult to eliminate: (1) the incapability to execute real-time estimation and (2) extra components is essential. In order to eliminate mentioned problems, we propose the dynamic impedance method and the projection technique respectively to evaluate the SOC and SOH of Li-ion battery modules.

First of all, the percentage of the voltage and current variations (ΔV , ΔI) of a battery module are defined as the dynamic impedance. We can obtain results in a short period

(about 10 seconds) by using the proposed method. Moreover, in order to simply calculate the SOH of the battery, we use the proposed projection method and the dynamic impedance results, and any extra testing devices or circuits are not essential. Actually, the used parameters are repeatedly updated during the calculation process depending on the SOH of the battery, thus reflecting the true SOC of the Li-ion battery. In this work, the related descriptions and theoretical predictions of the two proposed approaches are detail.

For SOC estimation we employ voltage and current variations ΔV and ΔI captured by the battery management system from the tested battery module under certain load with a fixed sampling rate of once per second as shown in Fig. 1.

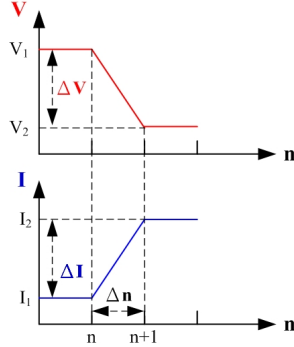


Fig. 1: Dynamic impedance ($\Delta V/\Delta I$).

$$\Delta V = |V_2 - V_1| \quad (1)$$

$$\Delta I = |I_2 - I_1| \quad (2)$$

The ratio of Eq. (1) and Eq. (2) can be defined as the proposed dynamic impedance:

$$\frac{\Delta V}{\Delta I} = \frac{|V_2 - V_1|}{|I_2 - I_1|} \quad (3)$$

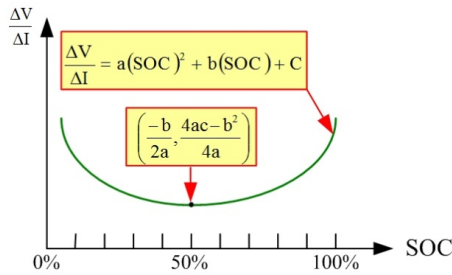


Fig. 2 Relationship of SOC versus dynamic impedance

A Li-ion battery module is tested to obtain the relationship between the SOC and the proposed dynamic impedance. All the measured information are recorded when the battery module are discharged from an SOC of 100 % to 0 % ; the simulated results are depicted in Fig. 2. This curve is like a parabolic feature. Therefore, the model of the dynamic impedance can be expressed as follows

$$\frac{\Delta V}{\Delta I} = a(\text{SOC})^2 + b(\text{SOC}) + C \quad (4)$$

Where the parameters a , b , and c reflect the features of the Li-ion battery. Equation (4) can be expressed as

$$\frac{\Delta V}{\Delta I} = a\left(\text{SOC} + \frac{b}{2a}\right)^2 + \left(\frac{4ac - b^2}{4a}\right) \quad (5)$$

From Eq. (5) we can obtain the lowest value of the curve at $(-b/2a, (4ac-b^2)/4a)$, where a is the curvature of the parabola. Another value determines the orientation of the parabola (downward if a is negative and upward if a is positive). Greater a value indicates larger curvature; therefore, when the battery is near being fully charged or empty, the dynamic impedance is higher. From experiments, we find that the aged batteries reveal greater voltage variations (ΔV) while they are near full or empty under the same load conditions, that is to say, the dynamic impedance is greater. From above-mention and Fig. 2, it can be presumed that the dynamic impedance of aged batteries is greater than that of new batteries when they are near empty or full. For this reason, the SOH of the battery can be determined through parameter a . Moreover, from Eq. (4), we can infer that one dynamic impedance value corresponds to two different SOC. To solve this issue, the rate of change in SOC is calculated to get a unique value: the only solution for the SOC can be obtained by using Eq. (4).

$$\frac{\Delta(\Delta V/\Delta I)}{\Delta \text{SOC}} = 2a(\text{SOC}) + b \quad (6)$$

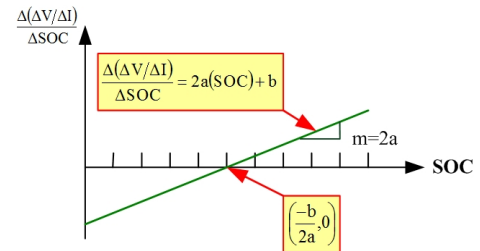


Fig. 3 Relationship between $\Delta(\Delta V/\Delta I)/\Delta \text{SOC}$ and SOC

According to Eq. (6), the relationship between $\Delta(\Delta V/\Delta I)/\Delta \text{SOC}$ and SOC can be sketched as shown in Fig. 3. This plot shows that the equation is an one-to-one function, and once we get $\Delta(\Delta V/\Delta I)/\Delta \text{SOC}$, the unique SOC of the battery can be obtained. If $\Delta(\Delta V/\Delta I)/\Delta \text{SOC}$ is zero, SOC is $-b/2a$. And then Eq. (6) can be expressed:

$$\text{SOC} = \left[\frac{\Delta(\Delta V/\Delta I)}{\Delta \text{SOC}} - b \right] / 2a \quad (7)$$

Equation (7) provides the unique solution of SOC. We can obtain $\Delta(\Delta V/\Delta I)$ for real-time calculation. When the discharge conditions of batteries vary, we can get a linear relationship between the discharging current and time. Fig. 4 shows the relationship between the current and the SOC.

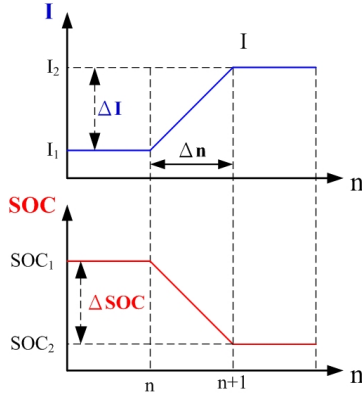


Fig. 4 The discharging current and SOC waveforms when the load changes.

It is apparent that ΔSOC can be obtained by using the coulomb counting method from Fig. 4:

$$\Delta SOC = |SOC_2 - SOC_1| = \frac{I/2 \times |I_2 - I_1| \times [(n+1) - n]}{Q_{rated}} = \frac{1}{2} \times \frac{\Delta I \times \Delta n}{Q_{rated}} = \frac{1}{2} \times \frac{\Delta I}{C_{rated} \times 3600} \quad (8)$$

In this study, we set the sampling rate of the BMS is once per second. Thus, $\Delta n=1$ s.

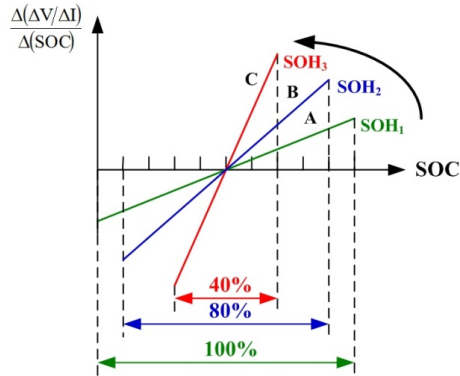


Fig. 5 $\Delta(\Delta V/\Delta I)/\Delta SOC$ vs. SOC with different SOH

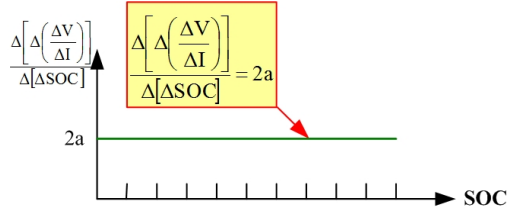


Fig. 6 Relationship curve between $\Delta(\Delta V/\Delta I)/\Delta SOC$ and SOC

Actually, from Fig. 2, we can infer that when the aging extent in Li-ion battery is more severe, the voltage increments and decrements are significant, and as a result, the dynamic impedance increases. In other words, we can obtain a greater slope of the SOC as shown in Fig. 5. Also, the projection length on the SOC axis is shorter. In Fig. 5, the curves A, B, and C respectively display the corresponding SOH to three aging degrees. Therefore, this study can provide a base for calculating the corresponding SOH by using the slope of the characteristic curve as shown in Fig. 5.

The discussion above indicates that by deriving parameters a and b , the SOC of the battery can be calculated

from Eq. (7). The parameters a and b can be derived by three steps as follows.

Step 1: Derive parameter a

First of all, the rate of change in $\Delta(\Delta V/\Delta I)/\Delta SOC$ versus the SOC can be used to depict the relationship between dynamic impedance and the SOC by computing two slopes, as shown in Fig. 6. Actually, it just need only derive the second derivative of the dynamic impedance versus the SOC to obtain a constant value $2a$. This can be expressed as

$$\frac{d^2(\Delta V/\Delta I)}{dSOC} = \frac{\Delta[\Delta(\Delta V/\Delta I)]}{\Delta[\Delta SOC]} = 2a \quad (9)$$

However, as shown in Fig. 5, the value of a is affected by the aging degree of the Li-ion battery, that is to say, a is not constant and changes continuously with the SOH of the battery.

Step 2: Determine parameter b

As shown in Fig. 3, the coordinates on the curve where

the rate of change is 0 are $\left(\frac{-b}{2a}, \frac{4ac-b^2}{4a}\right)$. Suppose these coordinates do not change with the SOH of the battery. Thus,

$$\left(\frac{-b}{2a}\right)_{now} = \left(\frac{-b}{2a}\right)_{origin} \quad (10)$$

where $(-b/2a)_{origin}$ is a reference value. This can be easily manipulated to obtain the value of parameter b .

$$b_{now} = 2a_{now} \times \left(\frac{b}{2a}\right)_{origin} \quad (11)$$

Similar to parameter a , the value of b always can be updated with the SOH of the Li-ion battery.

Step 3: Estimate SOH

As shown in Fig. 5, the slopes of the $\Delta(\Delta V/\Delta I)/\Delta SOC$ to SOC characteristic curves in oldish batteries are greater, and thus the curves present shorter projections on the SOC axis. This also indicates that the total amount of energy that the battery can store is less, and in other words, the SOH is poorer. Once the value of a is calculated in Step 1, we can use trigonometric functions to derive the current SOH of the battery, as depicted in Fig. 7; a_{origin} and a_{now} respectively denote the original and current slopes of the $\Delta(\Delta V/\Delta I)/\Delta SOC$ to SOC characteristic curves, the former denoting the parameter of a new battery; θ_{origin} and θ_{now} represent the angles between the curves and the SOC axis. From Fig. 7, we can deduce that the current SOH of the battery is

$$SOH_{now} = \frac{L \times \cos \theta_{now}}{L \times \cos \theta_{origin}} = \frac{\cos \theta_{now}}{\cos \theta_{origin}} \quad (12)$$

where L is the length of the characteristic curve. From the relationship between slope a_{origin} and angle θ_{origin} , we can derive that

$$a_{origin} = \frac{L \times \sin \theta_{origin}}{L \times \cos \theta_{origin}} = \tan \theta_{origin} \quad (13)$$

Using inverse trigonometric functions to compute Eq. (13), we can obtain $\theta_{origin} = \tan^{-1}(a_{origin})$ as well as $\theta_{now} = \tan^{-1}(a_{now})$.

By substituting these into Eq. (12), we can derive the SOH of the battery.

$$\text{SOH}_{\text{now}} = \frac{\cos(\tan^{-1} a_{\text{now}})}{\cos(\tan^{-1} a_{\text{origin}})} \quad (14)$$

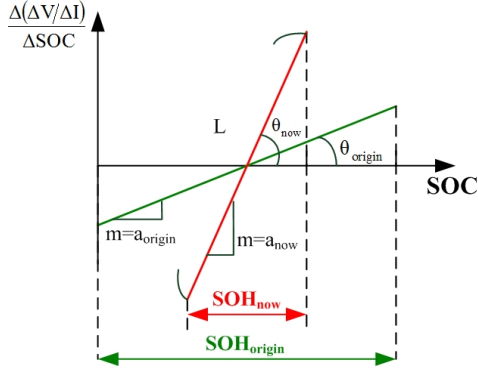


Fig. 7 Projection method for SOH estimation

III. ESTIMATION PROCEDURES OF SOC AND SOH

Fig. 8 displays the proposed procedure to estimate SOC and SOH. First, the BMS acquires the related information of the Li-ion battery modules, and transmits this information to the microcontroller by the I²C interface. When the dynamic impedance and parameters a_{now} and b_{now} are calculated, we can obtain the latest SOC and SOH.

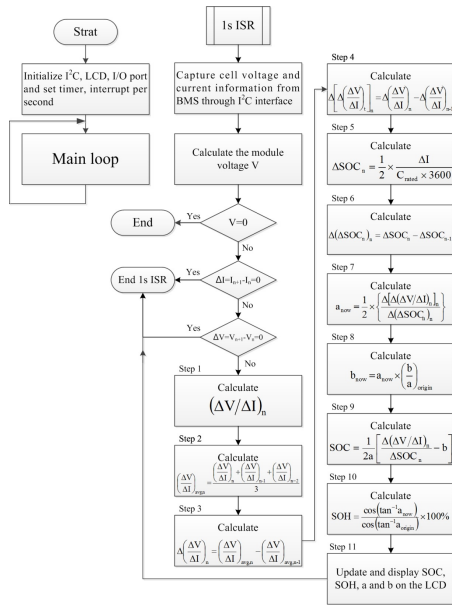


Fig. 8 The proposed estimation algorithm

The proposed estimation procedure shown in Fig. 8 is composed of a main program and an interrupt subroutine. Once the program starts, the I²C interface, and the related peripheral modules are initialized. In order to synchronize with the BMS, once system executes an interrupt each second, the timer is forced to enter interrupt service, that is to say, the procedure is executed repeatedly. Once procedure entrance into the interrupt subroutine, the BMS via

the I²C interface to retrieve the related information from battery module. And then, the battery module voltage V is captured. If V equals 0, then the protective functions is started, and the battery module is cutoff. Therefore, the proposed procedure is finished. And then, it is determined whether current change ΔI is 0. If it is, the interrupt is ended; on the other hand, the procedures continue. Then, if the voltage variation equals 0, the interrupt is finished; if not, then the SOC, and the SOH can be computed as shown in Fig. 8.

IV. EXPERIMENTAL RESULTS

In this study, we use the battery tester GBT-2636 as a verification experiment to validate the SOC and SOH estimation procedure. We can set the experiment parameters, including the discharge current amplitude, and the experiment duration. The used ICR18650-22P Li-ion batteries with 18650 packages are tested in our experiments. The related specifications are shown in Table 2.

Table 2 The specifications of the tested Li-ion battery

Rated capacity	Maximum charge current	Maximum voltage
2150 mAH	2.15 A(1 C)	4.2 V
Rated voltage	Maximum discharge current	Cut-off voltage
3.7 V	10 A(4.65 C)	2.75 V

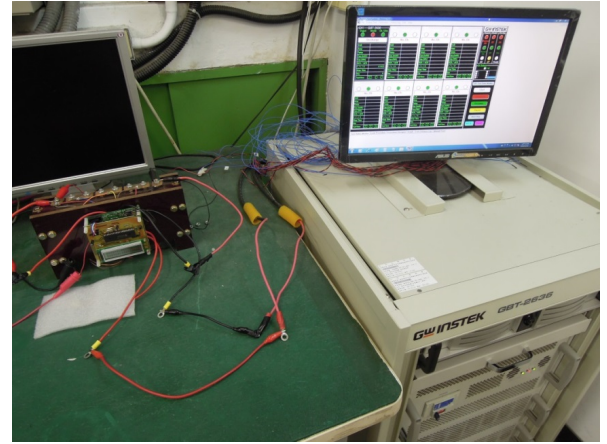


Fig. 9 The experiment setup of the SOC and SOH estimation

Fig. 9 presents the experimental setup, including the 8S1P battery module, the battery tester, and the implemented estimation platform. First of all, we programmed the discharge procedures by the GBT-2636, presetting the parameters of the experiments. The battery module is discharged by the GBT-2636 until the battery emptied, and the battery module voltage decreased to the cut-off voltage. All the related information are recorded by the GBT-2636 during the experiment. After analyzing above information, we can get ΔV , ΔI , and the dynamic impedance to plot the corresponding curves to note their trends and confirm the proposed estimation algorithm.

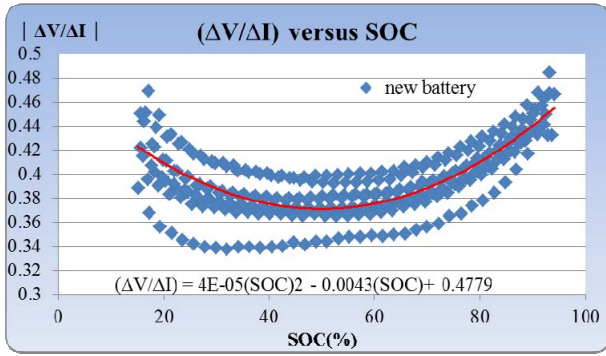


Fig. 10 Dynamic impedance versus SOC in new battery

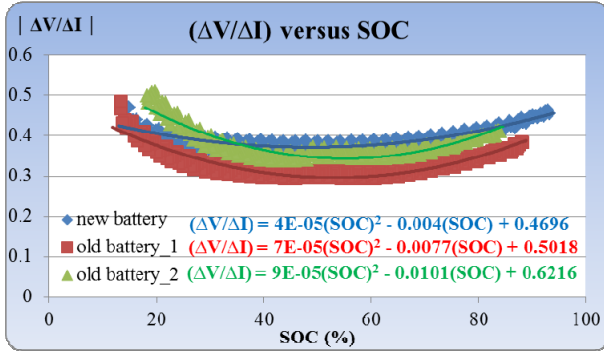


Fig. 11 Comparison of dynamic impedance versus SOC for new and aged batteries

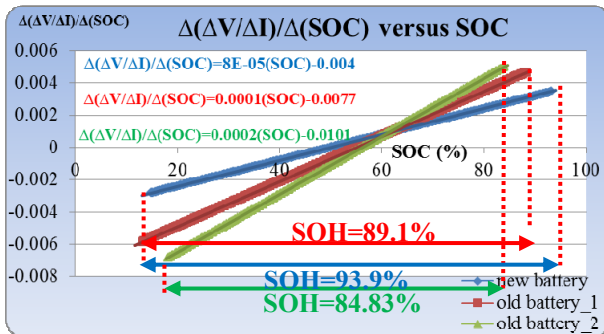


Fig. 12 Comparison of $\Delta(\Delta V/\Delta I)/\Delta SOC$ versus SOC for new and aged batteries

The characteristic curve which is calculated from the experimental data of a new battery is shown in Fig. 10. As the above-mentions, the dynamic impedance versus the SOC reveals a parabolic characteristic, where the red curve expresses the trend curve. Once the SOC is near 100 % or 0 %, the amplitude of the dynamic impedance is higher. By contrast, when the SOC is nearly 50 %, the dynamic impedance is lowest. In Fig. 10, the dynamic impedance and its corresponding SOC reveal a quadratic characteristic, that is to say, each dynamic impedance correlate with two values for SOC. Therefore, we can calculate the slope in the dynamic impedance with respect to the SOC to determine a single SOC value. The dynamic impedance versus SOC relationship of batteries with three different aging degrees is depicted as Fig. 11. It is apparent that the dynamic impedances of the aged batteries (old battery_1 and old

battery_2) are greater than the new battery, and the magnitudes of change in the SOC are also higher.

Next, we can obtain the relationships between $\Delta(\Delta V/\Delta I)/\Delta SOC$ and SOC (Fig. 12) by calculating the curves in Fig. 11. We can find that the slopes of the relationship for the aged batteries are greater and their corresponding projections on the SOC axis are shorter. That is to say, the aged batteries have smaller available capacities. In Fig. 12, it is clear that the available capacities of these two old one are about 85 % and 89 % of the rated capacity, respectively while of the new battery capacity reach about 94 %. From above experiments, the proposed algorithm can effectively evaluate the SOC and SOH of battery modules.

V. CONCLUSION

New evaluation methods for Li-ion battery module are proposed in this paper. In the first technique, the ratio between the variations in voltage and the variations in current in the charging and discharging operation of battery modules are defined to be the dynamic impedance, which further served as the base to estimate the SOC of the battery. Moreover, the proposed projection technique acquires the SOH by using the slope of the dynamic impedance versus SOC. The most important advantage of the proposed techniques is that initial values are not essential. Furthermore, we use simple operations to calculate various parameters and provide real time results for evaluating the SOC and SOH. In addition, the complete mathematical model of the SOC and SOH for the Li-ion battery module is derived in this work. All the theoretical predictions and experiments show that variations in the aging degree in the Li-ion battery module are reflected in the dynamic characteristics, which then change the parameters. Thus, the parameters update automatically with the aging degree of the battery module. In order to confirm the proposed evaluation algorithm for SOC and SOH, we implemented an estimation system without additional testing circuits that combines an MCU with the BMS. Finally, all the experimental results agree the theory predictions.

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